Jan Szczekulski

jszczekulski@ucsd.edu linkedin.com/in/jan-szczekulski github.com/szczekulskij

Summary

Software engineer with 5+ years designing and building distributed systems end to end, from warehouse associate-facing tools that reconcile physical and virtual item movement, to internal configuration frontends, high-throughput A/B testing backend integrations, and network-wide fulfillment services. I take pride in ownership, data-driven decisions, user obsession, and forward-thinking system design with clean interfaces within and across services.

Experience

Amazon — Seattle, USA

Software Development Engineer Apr 2025 – Present

- Designed and led the end-to-end deprecation of a 17-year-old Tier-1 configuration service (∼**10M calls/day**; a dependency for 5 fulfillment workflows across 7 systems), re-architecting configuration data from 9 scattered database tables into 3 clean DynamoDB-backed APIs. Delivered as 93 code reviews and 12 coordinated production deployments with **zero site impact**, removing ∼9k lines of parsing code logic and closing a service-authentication security gap.
- Designed and built a backend service sitting in the request path of a core fulfillment platform, engineered for **10k requests/second**; making design tradeoffs between scalability, simplicity, externalization and robustness in terms of infrastructure and low-level development detail.
- Designed and extended a core outbound-flow routing system to account for upstream machines capacity and physical graph-like connectivity of underlying process paths. Designed and implemented a new Linear Programming optimization objective which limits the load of process path based on upstreams' fullness signal (use case entitlement ∼**$14.3M** annualized).
- Translated live site-reported bugs into clean designs and implementations, resulting in warehouse-automation features with measurable impact: a bulk-item handling workflow now live at **50+ sites** (part of a **$45.6M/yr** program), an auto-login capability removing ∼3 s from every associate login worldwide, and an automatic mis-sort detection system eliminating ∼**10k** mis-routed containers per month.
- Root-caused and **halved latency (p50/p99)** across 10+ APIs of a Tier-1 item sortation service that was overloading during peak season. Designed and built the missing observability metrics, removed redundant service calls, and parallelized hot paths, ending a multi-week run of latency incidents (e.g. core API 600 ms → 400 ms; item-lookup API 420 ms → 230 ms).
- Owned 6+ Tier-1 production systems end-to-end (on-call, incident response, load-test and scaling-readiness authoring, migrations); led 12 peak-preparedness meetings, deep-diving and sharing information on possible system risks, and available low-level levers and actionable responses to avoid peak severity events, resulting in no system downtime.
- Designed and built distributed backend services in **Java/Spring** on **DynamoDB, ECS, Lambda, SQS, SNS, AppConfig and Kinesis**, including large-scale migration infrastructure (double-read/double-write validation, staged traffic shifting, rollback mechanisms) and self-service operations tooling that cut site-configuration changes from ∼2 days to ∼15 minutes.

UCSD — San Diego, USA

Graduate Research Assistant @ Contextual Robotics Institute Sept 2024 – March 2025
Teaching Assistant Jan 2025 – March 2025

- Co-author and lead implementer of a state-of-the-art robotic tabletop rearrangement system published at **ICRA 2025**, focusing on long-horizon planning and robust mechanical execution under physical-world's uncertainty.
- Built planning systems integrating vision-language models and large language models for mobile manipulation and multi-step reasoning tasks.
- As a TA, I helped prepare study resources and provided office hours as well as seminars for a Search and Optimization course

The Hut Group (THG) — Manchester, UK

Software Engineer Dec 2021 – Aug 2023
Data Scientist Sept 2020 – Dec 2021

- Designed and implemented experimentation infrastructure integrating an in-house A/B testing backend with a high-throughput widget delivery platform serving **5k+ RPS**.
- Built backend APIs, integration services, rollout tooling, and React-based frontend systems enabling rapid experiment configuration supporting real-time configuration updates, feature exposure logic, and analytics collection in large-scale e-commerce systems contributing to **£150M+ YoY business impact**.
- Designed and implemented backend integrations between experimentation systems, widgeting services, and analytics platforms using **TypeScript, Python, SQL, React.js, and MVC-based architectures**.
- Built SQL and Python pipelines for experiment evaluation, statistical analysis, and operational analytics across millions of daily events.
- Managed production infrastructure including Kubernetes clusters, deployment pipelines, VM provisioning, patching, and operational maintenance.

Publications

OmniManip: Towards General Robotic Manipulation via Object-Centric Interaction Primitives as Spatial Constraints
ICRA 2025
– Co-author; contributed to long-horizon robotic planning, manipulation systems, and LLM-assisted embodied AI reasoning.

In Search for Optimal Schedule During Long-Term Treatment of Port-Wine Birthmarks
Scientific Reports
– Co-author; applied machine learning and statistical analysis methods to medical outcome prediction from imaging datasets.

Education

University of California, San Diego — San Diego, USA
M.S. in Computer Science, GPA: 3.85 2023 – 2025

University of Liverpool — Liverpool, UK
B.Sc. (Hons) Computer Science & Mathematics, First Class 2017 – 2020

Technical Skills

Data: Extensive Data Analysis experience, A/B testing, Linear Programming, Statistical Analysis, Data Aggregation Systems

Backend & Distributed Systems: Distributed Systems, Load balancing and scaling, Data Migration, Reliability Engineering, API Design, High-Throughput Systems

ML & Research: Agentic development, Robotics Planning Systems, PyTorch, Vision-Language Models, Large Language Models

Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes

Languages: Java, Go, Python, C, TypeScript, SQL

Frontend: React.js, javascript, MVC Architectures